Proofs of Lemma 3 and Theorem 2 in "Dory: Streaming PCG with Small Memory"

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1 Background on H-Coefficient Technique

H-coefficient technique [Pat09, CS14] can be used to bound the advantage of an information-theoretic adversary. We borrow the summary of the H-coefficient technique from [GKWY20]. Let $\mathcal{T}$ be the set of *attainable* transcripts for some information-theoretic distinguisher, and let $\Pr_{\mathsf{real}}[\cdot]$ and $\Pr_{\mathsf{ideal}}[\cdot]$ be the probabilities in the real and ideal worlds, respectively. The H-coefficient technique divides $\mathcal{T}$ into a "bad" subset $\mathcal{T}_{\mathsf{bad}}$ and a "good" subset $\mathcal{T}_{\mathsf{good}} := \mathcal{T} \setminus \mathcal{T}_{\mathsf{bad}}$, and shows that if it holds that

$$\Pr_{\mathsf{ideal}}[\mathcal{Q} \in \mathcal{T}_{\mathsf{bad}}] \leq \epsilon_1,$$

$$\forall \mathcal{Q} \in \mathcal{T}_{\mathsf{good}} : \frac{\Pr_{\mathsf{real}}[\mathcal{Q}]}{\Pr_{\mathsf{ideal}}[\mathcal{Q}]} \geq 1 - \epsilon_2,$$

the advantage of the distinguisher is at most $\epsilon_1 + \epsilon_2$.

2 Proof of Lemma 3

Lemma 3. *In the $(\mathcal{F}_{\mathsf{sVOLE}}, \mathsf{RO})$-hybrid model, the protocol $\Pi_{\mathsf{sPCG\text{-}sVOLE}}$ (Fig. 5) UC-realizes the functionality $\mathcal{F}_{\mathsf{sPCG\text{-}sVOLE}}$ (Fig. 4) against any malicious receiver P_1 if hash function $\mathsf{H}(x) = \pi(\sigma(x)) + \sigma(x)$, where $\pi : \mathbb{K} \to \mathbb{K}$ is a (programmable) random permutation and $\sigma : \mathbb{K} \to \mathbb{K}$ is an efficiently computable linear orthomorphism.*

Proof. When the receiver is corrupted, the view of the malicious adversary includes $\{\boldsymbol{c}_i\}_i$, ϕ, and the outputs of $\mathcal{F}_{\mathsf{sVOLE}}$ and RO (and the random permutation for H calls in cGGM). In the simulation, the simulator simply sends uniformly random $\{\boldsymbol{c}_i\}_i$ and ϕ to the corrupted P_1 and emulates RO and the random permutation on-the-fly. Moreover, extended from the simulation strategy of [GYW+23, Theorem 7, full version] against the corrupted receiver, the simulator observes each query of the adversary to RO or the random permutation to compute a guess Δ', sends Δ' to the functionality to check whether $\Delta' = \Delta$, and programs RO and the random permutation to be consistent with $(\{\boldsymbol{c}_i\}_i, \phi)$ and the correct guess Δ. We can bound the advantage of the adversary via the H-coefficient technique [Pat09, CS14, DLMS14] as in [GYW+23]. This advantage is still negligible. To make this work self-contained, we provide a detailed proof as follows.

When the receiver is corrupted, the simulator $\mathcal{S}$ in the ideal world emulates the random permutation π and its inverse π^{-1} throughout the lifespan of the distinguishing game. During the distinguishing game in either world, let $\mathcal{Q}_\pi$ denote an *ordered* list of the adversary's query-answer pairs of the random permutation or its inverse (note that $(x, y) \in \mathcal{Q}_\pi$ means that the adversary learns $\pi(x) = y$, regardless of whether it queried $\pi(x)$ or $\pi^{-1}(y)$), and $\mathcal{Q}_{\mathcal{O}}$ denote an *ordered* list of the adversary's query-answer pairs of the "oracle" emulated by the non-corrupted machine(s) (i.e., the honest P_0 plus the subroutine $\mathcal{F}_{\mathsf{sVOLE}}$ in the real world, or the simulator $\mathcal{S}$ in the ideal world). One can think that, for each $i \in [t]$ execution in the **Initialize** phase, $\mathcal{Q}_{\mathcal{O}}$ records the query $(\beta_i, \mathsf{M}(\beta_i), \boldsymbol{r}_i, \mathsf{M}(\boldsymbol{r}_i))$ (i.e., the effective sVOLE output committed by the corrupted P_1, see below) with its answer $\boldsymbol{c}_i$ (i.e., the transcript sent by the honest P_0). $\mathcal{S}$ has access to $\mathcal{Q}_\pi$ and $\mathcal{Q}_{\mathcal{O}}$ in the ideal world.

$\mathcal{S}$ sends (init) to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ upon receiving the first (init) from $\mathcal{A}$ to $\mathcal{F}_{\mathsf{sVOLE}}$, and works as follows for each $i \in [t]$ parallel execution in the **Initialize** phase:

1. Upon receiving $(\mathsf{extend}, h+1)$ from $\mathcal{A}$ to $\mathcal{F}_{\mathsf{sVOLE}}$, $\mathcal{S}$ waits for $\mathcal{A}$ to choose its sVOLE output $(\boldsymbol{s}_i, \mathsf{M}(\boldsymbol{s}_i))$.
2. $\mathcal{S}$ extracts $(\beta_i, \mathsf{M}(\beta_i))$, receives $\boldsymbol{d}_i$ from $\mathcal{A}$, and
   (a) If $\beta_i = 0$, picks a random $\boldsymbol{r}_i$.
   (b) If $\beta_i \neq 0$, extracts $\boldsymbol{r}_i := \beta_i^{-1} \cdot (\boldsymbol{d}_i - \boldsymbol{s}_i{}^{(1,h)})$.
3. $\mathcal{S}$ computes $\mathsf{M}(\boldsymbol{r}_i) := \boldsymbol{r}_i \cdot \mathsf{M}(\beta_i) - \mathsf{M}(\boldsymbol{s}_i{}^{(1,h)})$.
4. $\mathcal{S}$ sends random $\boldsymbol{c}_i$ to $\mathcal{A}$ and appends $((\beta_i, \mathsf{M}(\beta_i), \boldsymbol{r}_i, \mathsf{M}(\boldsymbol{r}_i)), \boldsymbol{c}_i)$ to $\mathcal{Q}_{\mathcal{O}}$.

In the consistency check in the **Initialize** phase:

1. Upon receiving (extend, μ) from $\mathcal{A}$ to $\mathcal{F}_{\mathsf{sVOLE}}$, $\mathcal{S}$ waits for $\mathcal{A}$ to choose its sVOLE output $(\boldsymbol{m}, \mathsf{M}(\boldsymbol{m}))$.
2. $\mathcal{S}$ computes $(\boldsymbol{e}, \mathsf{M}(\boldsymbol{e}))$ as an honest P_1 in the protocol and receives $(\mathsf{U}, \boldsymbol{z})$ from $\mathcal{A}$.
3. $\mathcal{S}$ computes $(\boldsymbol{z}', w)$ as an honest P_1 in the protocol. If $\boldsymbol{z}' \neq \boldsymbol{z}$, $\mathcal{S}$ sends random ϕ to $\mathcal{A}$; otherwise, $\mathcal{S}$ emulates the random oracle and sends $\mathsf{RO}(w)$ to $\mathcal{A}$.

In the **Extend** phase:

1. $\mathcal{S}$ uses the same randomness to sample $\boldsymbol{b}$ and computes $(x, \mathsf{M}(x))$ as an honest P_1 in the protocol.
2. $\mathcal{S}$ sends (extend) and $(x, \mathsf{M}(x))$ to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$.

Moreover, $\mathcal{S}$ invokes **Global-key queries** to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ in the following cases (where, in $\mathcal{Q}_{\mathcal{O}}$, we omit the subscript of the parallel executions for clarity):

- query_1: $\mathcal{Z}$ queries the random permutation π with x or its inverse π^{-1} with y. For each $((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}) \in \mathcal{Q}_{\mathcal{O}}$ where $\beta \neq 0$:
  1. $\mathcal{S}$ computes
  $$\begin{aligned} \alpha = \alpha_0 \dots \alpha_{h-1} &:= \overline{\boldsymbol{r}}^{(0)} \dots \overline{\boldsymbol{r}}^{(h-1)}, \\ \{z_i\}_{i\in[h]} &:= \mathsf{cGGM.OffPath}(\alpha, \{\mathsf{sum}^{(i)}_{\overline{\alpha}_i}\}_{i\in[h]}), \\ \forall j \in [1, h-1] : w_j &:= \textstyle\sum_{i\in[j]} z_i, \end{aligned} \tag{1}$$
  where $\mathsf{cGGM.OffPath}$ is a macro such that, on input a path $\alpha \in \{0,1\}^h$ and h sums $\{\mathsf{sum}^{(i)}_{\overline{\alpha}_i}\}_{i\in[h]}$ used in $\mathsf{cGGM.PuncFullEval}$ to define an h-level correlated GGM tree except the h on-path nodes, it outputs the h off-path nodes for the path α. This macro ensures that, for $j \in [h]$,
  $$z_j = \mathsf{sum}^{(j)}_{\overline{\alpha}_j} - \text{“other } \overline{\alpha}_j\text{-side nodes on the } j\text{-th level} \\ \text{defined by the off-path nodes } \{z_i\}_{i\in[j]}\text{”}.$$
  2. For each $j \in [1, h-1]$, $\mathcal{S}$ extracts $\mathsf{K}(\beta)'_1 \in \mathbb{K}$ and sends $(\mathsf{guess}, \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta)'_1))$ to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ if $\mathcal{Z}$ queries π with x, or extracts $\mathsf{K}(\beta)'_2 \in \mathbb{K}$ and $(\mathsf{guess}, \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta)'_2))$ to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ if $\mathcal{Z}$ queries π^{-1} with y, where
  $$\begin{aligned} \mathsf{K}(\beta)'_1 &:= \sigma^{-1}(x) + w_j, \\ \mathsf{K}(\beta)'_2 &:= \begin{cases} \sigma^{-1}(-y + z_j) + w_j & \text{, if } \overline{\alpha}_j = 0 \\ \sigma'^{-1}(-y - z_j) + w_j & \text{, if } \overline{\alpha}_j = 1 \end{cases} \end{aligned} \tag{2}$$
  Note that $\sigma'(x) := \sigma(x) - x$ is a permutation as $\sigma : \mathbb{K} \to \mathbb{K}$ is an orthomorphism, and its inverse σ'^{-1} should be well-defined.

- query_2: A new pair $((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c})$ where $\beta \neq 0$ is added to $\mathcal{Q}_{\mathcal{O}}$ during execution. Using this pair, $\mathcal{S}$ computes $\{z_i\}_{i\in[h]}$ and $\{w_j\}_{j\in[1,h-1]}$ as per (1). Then, for each $(x, y) \in \mathcal{Q}_\pi$ and each $j \in [h]$, $\mathcal{S}$ extracts both $\mathsf{K}(\beta)'_1$ and $\mathsf{K}(\beta)'_2$ as above, and sends two global-key guesses $(\mathsf{guess}, \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta)'_1))$ and $(\mathsf{guess}, \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta)'_2))$ to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$.
- query_3: $\mathcal{Z}$ queries the random oracle RO with x. If $\boldsymbol{z}' \neq \boldsymbol{z}$, $\mathcal{S}$ computes

$$\Delta_x := (\textstyle\sum_{i\in[\mu]}(\boldsymbol{z}'^{(i)} - \boldsymbol{z}^{(i)}) \cdot \mathsf{X}^i)^{-1} \cdot (x - w),$$

and sends $(\mathsf{guess}, \Delta_x)$ to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$.

In query_1 and query_2, if $\mathcal{S}$ receives $(\mathsf{success})$ from $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ for some guess Δ, $\mathcal{S}$ will program the random permutation π and its inverse π^{-1} such that, for the offset $\mathsf{K}(\beta) := \mathsf{M}(\beta) - \beta \cdot \Delta$, the $\{z_i\}_{i\in[h]}$ and $\{w_j\}_{j\in[1,h-1]}$ computed from each pair in $\mathcal{Q}_{\mathcal{O}}$ up to this time, and each $j \in [1, h-1]$,

$$\pi(\sigma(\mathsf{K}(\beta) - w_j)) = -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j), \tag{3}$$

which must hold in the real world due to the construction of H and the definition of z_j and w_j. That is, the programming artificially ensures the real-world consistency among the off-path nodes on the cGGM tree in the ideal world.

In query_3, if $\mathcal{S}$ receives $(\mathsf{success})$ from $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ for some guess Δ, $\mathcal{S}$ will program the random oracle RO such that $\mathsf{RO}(x) = \phi$. This case is to ensure that $\mathcal{Z}$ cannot use its known Δ and $\boldsymbol{z}' \neq \boldsymbol{z}$ to distinguish the two worlds by querying RO only. The simulation in this case is perfect and will not affect the following statistical indistinguishability.

We use H-coefficient technique to prove that this ideal world is statistically indistinguishable from the real one. Without loss of generality, we consider a deterministic environment $\mathcal{Z}$ and the transcript $\mathcal{Q}$, which is a part of the joint distribution of the view of $\mathcal{A}$ and the output of the honest P_0 (note that the remaining part of this joint distribution is the $\mathsf{K}(x)$'s in all **Extend** executions, and its consistency with $\mathcal{Q}$ will be checked in the end).

A transcript $\mathcal{Q} = (\Delta, \mathcal{Q}_\pi, \mathcal{Q}_{\mathcal{O}})$ is **bad** if it falls into one of the following cases:

- bad_1. There exist two distinct transcript pairs

$$\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1],$$
$$\Big(((\beta', \mathsf{M}(\beta'), \boldsymbol{r}', \mathsf{M}(\boldsymbol{r}')), \boldsymbol{c}'), j'\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$$

such that, for $\mathsf{K}(\beta) := \mathsf{M}(\beta) - \beta \cdot \Delta$ and $\mathsf{K}(\beta') := \mathsf{M}(\beta') - \beta' \cdot \Delta$, it holds that

$$\begin{aligned} &\Big(\mathsf{M}(\beta) - \beta \cdot \Delta - w_j = \mathsf{M}(\beta') - \beta' \cdot \Delta - w'_{j'}\Big) \\ \vee &\begin{pmatrix} -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j) \\ = -\sigma(\mathsf{K}(\beta') - w'_{j'}) + (-1)^{\overline{\alpha}'_{j'}} \cdot (-\overline{\alpha}'_{j'} \cdot (\mathsf{K}(\beta') - w'_{j'}) + z'_{j'}) \end{pmatrix}, \end{aligned}$$

where the values are defined as per (1). This bad case captures the collision between the queries to the random permutation and its inverse.

- bad_2. There exists a transcript pair

$$\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$$

such that, for $\mathsf{K}(\beta) := \mathsf{M}(\beta) - \beta \cdot \Delta$, it holds that

$$\begin{aligned} &\begin{pmatrix} (\sigma(\mathsf{K}(\beta) - w_j), \dots) \in \mathcal{Q}_\pi \\ \vee\, (\dots, -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)) \in \mathcal{Q}_\pi \end{pmatrix} \\ &\wedge \Big(\pi(\sigma(\mathsf{K}(\beta) - w_j)) \neq -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)\Big), \end{aligned}$$

where the values are defined as per (1). This bad case captures the inconsistency in the already defined entries of the random permutation and its inverse.

Consider bad_1 in the ideal world. For any two distinct pairs in $\mathcal{Q}_{\mathcal{O}} \times [1, h-1]$,

$$\begin{aligned}
&\Pr_{\mathsf{ideal}}\left[\begin{array}{r}\left(\mathsf{M}(\beta) - \beta \cdot \Delta - w_j = \mathsf{M}(\beta') - \beta' \cdot \Delta - w'_{j'}\right)\\ \vee\left(\begin{array}{l} -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j) \\ \qquad = -\sigma(\mathsf{K}(\beta') - w'_{j'}) + (-1)^{\overline{\alpha}'_{j'}} \cdot (-\overline{\alpha}'_{j'} \cdot (\mathsf{K}(\beta') - w'_{j'}) + z'_{j'})\end{array}\right)\end{array}\right] \\
&\leq \Pr_{\mathsf{ideal}}\left[\mathsf{M}(\beta) - \beta \cdot \Delta - w_j = \mathsf{M}(\beta') - \beta' \cdot \Delta - w'_{j'}\right] \\
&\quad + \Pr_{\mathsf{ideal}}\left[\begin{array}{l} -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j) \\ \qquad = -\sigma(\mathsf{K}(\beta') - w'_{j'}) + (-1)^{\overline{\alpha}'_{j'}} \cdot (-\overline{\alpha}'_{j'} \cdot (\mathsf{K}(\beta') - w'_{j'}) + z'_{j'})\end{array}\right] \\
&= \frac{1}{|\mathbb{K}|} + \frac{1}{|\mathbb{K}|} = \frac{2}{|\mathbb{K}|},
\end{aligned}$$

where the probabilities come from that (i) the value of $\boldsymbol{c}$ in $\mathcal{Q}_{\mathcal{O}}$ is uniform and pairwise independent in the ideal world, and (ii) each z_j computed from $\boldsymbol{c}$ is also uniform due to (1). Taking a union bound over all distinct pairs over $\mathcal{Q}_{\mathcal{O}} \times [1, h-1]$, it holds that

$$\Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1\right] \leq \frac{t^2(h-1)^2}{|\mathbb{K}|}. \tag{4}$$

As for bad_2, it is sufficient to bound $\Pr_{\mathsf{ideal}}\left[\mathsf{bad}_2 \;\middle|\; \neg\, \mathsf{bad}_1\right]$ using the following three sub-cases. In the *first* sub-case where

$$\begin{aligned}
&\forall\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1] : \\
&\quad\left(\begin{array}{r} (\sigma(\mathsf{K}(\beta) - w_j), \dots) \notin \mathcal{Q}_\pi \\ \wedge\, (\dots, -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)) \notin \mathcal{Q}_\pi \end{array}\right),
\end{aligned}$$

it is clear that bad_2 will not happen.

Otherwise, in the following two complement sub-cases, $\mathcal{S}$ can extract the global key Δ (which is a part of the transcript $\mathcal{Q}$) by solving $\mathsf{K}(\beta)$ from either $x = \sigma(\mathsf{K}(\beta) - w_j)$ or $y = -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)$ according to (2) and computing $\Delta = \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta))$ for each pair in $\mathcal{Q}_{\mathcal{O}}$ with $\beta \neq 0$. Then, $\mathcal{S}$ will use Δ to program π and π^{-1} to keep (3). If the programming succeeds, bad_2 will not happen since (3) holds for each pair in $\mathcal{Q}_{\mathcal{O}}$ and each $j \in [1, h-1]$.

In the *second* sub-case where a new query to π or π^{-1} triggers the programming (i.e., query_1 case), the programming must succeed conditioned on $\neg\mathsf{bad}_1$. The condition $\neg\mathsf{bad}_1$ ensures that, for every $\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$, the pre-image $x := \sigma(\mathsf{K}(\beta) - w_j)$ and the image $y := -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)$ to be programmed do not incur collision. More importantly, these two entries are not determined by the query/answer pairs in $\mathcal{Q}_\pi$ up to this time. Otherwise, $\exists(x, \dots) \in \mathcal{Q}_\pi$ or $\exists(\dots, y) \in \mathcal{Q}_\pi$ ahead of the current query such that the programming must have happened for the same $\mathcal{Q}_{\mathcal{O}}$ (i.e., contradicting that the programming is triggered by the current query). Hence, all pairs in $\mathcal{Q}_{\mathcal{O}}$ can be programmed as per (3). The answer to the current query is determined by the programmed result. The surely successful programming makes bad_2 impossible.

Consider the *third* sub-case where the programming arises from a new pair added to $\mathcal{Q}_{\mathcal{O}}$ (i.e., query_2 case). Assume that the global key Δ (and its associated $\mathsf{K}(\beta) = \mathsf{M}(\beta) - \beta \cdot \Delta$) to be used in the programming is extracted from this pair, an existing $(x^*, \dots) \in \mathcal{Q}_\pi$ or $(\dots, y^*) \in \mathcal{Q}_\pi$, and $j^* \in [1, h-1]$. Note that either $(x^*, \dots) \in \mathcal{Q}_\pi$ or $(\dots, y^*) \in \mathcal{Q}_\pi$ has been given to $\mathcal{Z}$ and *cannot* be programmed. We bound the probability of this *undesired* sub-case. In the ideal world,

$$\begin{aligned}
&\Pr_{\mathsf{ideal}}\left[\begin{array}{r}(x^* = \sigma(\mathsf{K}(\beta) - w_{j^*}))\\ \vee\, (y^* = -\sigma(\mathsf{K}(\beta) - w_{j^*}) + (-1)^{\overline{\alpha}_{j^*}} \cdot (-\overline{\alpha}_{j^*} \cdot (\mathsf{K}(\beta) - w_{j^*}) + z_{j^*}))\end{array}\right]\\
&\leq \Pr_{\mathsf{ideal}}\left[x^* = \sigma(\mathsf{K}(\beta) - w_{j^*})\right]\\
&\qquad + \Pr_{\mathsf{ideal}}\left[y^* = -\sigma(\mathsf{K}(\beta) - w_{j^*}) + (-1)^{\overline{\alpha}_{j^*}} \cdot (-\overline{\alpha}_{j^*} \cdot (\mathsf{K}(\beta) - w_{j^*}) + z_{j^*})\right]\\
&= \Pr_{\mathsf{ideal}}\left[w_{j^*} = \mathsf{K}(\beta) - \sigma^{-1}(x^*)\right]\\
&\qquad + \Pr_{\mathsf{ideal}}\left[z_{j^*} = (-1)^{\overline{\alpha}_{j^*}} \cdot \Big(y^* + \sigma(\mathsf{K}(\beta) - w_{j^*})\Big) + \overline{\alpha}_{j^*} \cdot (\mathsf{K}(\beta) - w_{j^*})\right]\\
&= \frac{1}{|\mathbb{K}|} + \frac{1}{|\mathbb{K}|} = \frac{2}{|\mathbb{K}|},
\end{aligned}$$

where the values are computed from the newly added pair as per (1), and the probabilities are taken over the uniform $\boldsymbol{c}$ in $\mathcal{Q}_{\mathcal{O}}$. Taking a union bound over $\mathcal{Q}_\pi \times \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$ (where we assume $|\mathcal{Q}_\pi| = p = p(\lambda)$ is polynomially bounded), we can see that this sub-case happens with probability at most $2pt(h-1)/|\mathbb{K}|$.

Combining the above three sub-cases, we have that

$$\Pr_{\mathsf{ideal}}\left[\mathsf{bad}_2 \,\middle|\, \neg\,\mathsf{bad}_1\right] \leq \frac{2pt(h-1)}{|\mathbb{K}|}. \tag{5}$$

Using (4) and (5), it holds that

$$\begin{aligned}
\Pr_{\mathsf{ideal}}\left[\mathcal{Q} \in \mathcal{T}_{\mathsf{bad}}\right] &= \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1 \vee \mathsf{bad}_2\right]\\
&= \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1\right] + \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_2\right] - \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1 \wedge \mathsf{bad}_2\right]\\
&= \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1\right] + \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_2 \,\middle|\, \neg\,\mathsf{bad}_1\right] \cdot \Pr_{\mathsf{ideal}}\left[\neg\mathsf{bad}_1\right]\\
&\leq \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1\right] + \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_2 \,\middle|\, \neg\,\mathsf{bad}_1\right]\\
&\leq \frac{t^2(h-1)^2 + 2pt(h-1)}{|\mathbb{K}|} = \mathsf{negl}(\lambda).
\end{aligned} \tag{6}$$

We proceed to bound the ratio of $\Pr_{\mathsf{real}}\left[\mathcal{Q}\right]$ to $\Pr_{\mathsf{ideal}}\left[\mathcal{Q}\right]$ for some fixed **good** transcript $\mathcal{Q} = (\Delta, \mathcal{Q}_\pi, \mathcal{Q}_{\mathcal{O}})$. Let $\mathcal{L} \subseteq \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$ such that

$$\begin{aligned}
&\forall\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{L} :\\
&\left(\begin{array}{r}(\sigma(\mathsf{K}(\beta) - w_j), \dots) \in \mathcal{Q}_\pi\\ \vee\, (\dots, -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)) \in \mathcal{Q}_\pi\end{array}\right).
\end{aligned}$$

Conditioned on $\neg(\mathsf{bad}_1 \vee \mathsf{bad}_2)$ implicit in good transcripts, $\mathcal{L}$ induces a subset $\mathcal{Q}_\pi(\mathcal{L}) \subseteq \mathcal{Q}_\pi$. There exists a *bijective* correspondence between a $(x, y) \in \mathcal{Q}_\pi(\mathcal{L})$ and the (z_j, w_j) computed as per (1) from a pair in $\mathcal{L}$ such that

$$\begin{aligned}
x &= \sigma(\mathsf{K}(\beta) - w_j),\\
y &= -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j).
\end{aligned}$$

It is clear that $|\mathcal{L}| = |\mathcal{Q}_\pi(\mathcal{L})| \leq \min(|\mathcal{Q}_\pi|, |\mathcal{Q}_{\mathcal{O}}| \cdot (h-1)) = \min(p, t(h-1))$.

For $\mathcal{X} \subseteq \mathcal{Q}_\pi$, let $\pi \sim \mathcal{X}$ be the event that the permutation π is consistent with the queries and answers in $\mathcal{X}$, i.e., $\forall (x, y) \in \mathcal{X} : \pi(x) = y$. Let $\boldsymbol{c}(j) \sim \mathcal{Q}_{\mathcal{O}}$ be the event that the prefix $({\boldsymbol{c}_i}^{(0)}, \dots, {\boldsymbol{c}_i}^{(j-1)})$ in each $i \in [t]$ parallel execution in the **Initialize** phase take the literal values in $\mathcal{Q}_{\mathcal{O}}$. For $\mathcal{Y} \subseteq \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$, let $(\pi, \Delta) \sim \mathcal{Y}$ be the *real-world* event that π and the global key Δ are consistent with the pairs in $\mathcal{Y}$, i.e.,

$$\begin{aligned}
&\forall\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Y} :\\
&\quad \pi(\sigma(\mathsf{K}(\beta) - w_j)) = -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)
\end{aligned}$$

for z_j and w_j computed from (1). Let $(a)_b := a \cdots (a-b+1)$ be falling factorial.

In the real world, $\pi : \mathbb{K} \to \mathbb{K}$ is a random permutation without programming, and

$$\begin{aligned}
\Pr_{\mathsf{real}}[\mathcal{Q}] &= \Pr_{\mathsf{real}}\left[\boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}} \wedge (\pi, \Delta) \sim \mathcal{Q}_{\mathcal{O}} \times [1, h-1] \,\middle|\, \pi \sim \mathcal{Q}_\pi \wedge \Delta\right] \\
&\quad \cdot \Pr_{\mathsf{real}}[\pi \sim \mathcal{Q}_\pi \wedge \Delta] \\
&= \Pr_{\mathsf{real}}\left[\boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}} \wedge (\pi, \Delta) \sim \mathcal{Q}_{\mathcal{O}} \times [1, h-1] \,\middle|\, \pi \sim \mathcal{Q}_\pi \wedge \Delta\right] \\
&\quad \cdot \Pr_{\mathsf{real}}[\pi \sim \mathcal{Q}_\pi] \cdot \Pr_{\mathsf{real}}[\Delta] \\
&= \Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim \mathcal{Q}_{\mathcal{O}} \times [1, h-1] \,\middle|\, \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}} \end{array}\right] \\
&\quad \cdot \Pr_{\mathsf{real}}\left[\boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}} \,\middle|\, \pi \sim \mathcal{Q}_\pi \wedge \Delta\right] \cdot \Pr_{\mathsf{real}}[\pi \sim \mathcal{Q}_\pi] \cdot \Pr_{\mathsf{real}}[\Delta] \\
&= \Pr_{\mathsf{real}}\left[\begin{array}{c} (\pi, \Delta) \sim \mathcal{L} \\ \wedge\, (\pi, \Delta) \sim (\mathcal{Q}_{\mathcal{O}} \times [1, h-1]) \setminus \mathcal{L} \end{array} \,\middle|\, \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}} \end{array}\right] \\
&\quad \cdot \Pr_{\mathsf{real}}[\boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}}] \cdot \Pr_{\mathsf{real}}[\pi \sim \mathcal{Q}_\pi] \cdot \Pr_{\mathsf{real}}[\Delta] \\
&= \Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim (\mathcal{Q}_{\mathcal{O}} \times [1, h-1]) \setminus \mathcal{L} \,\middle|\, \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}} \wedge (\pi, \Delta) \sim \mathcal{L} \end{array}\right] \\
&\quad \cdot \Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim \mathcal{L} \,\middle|\, \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}} \end{array}\right] \cdot \frac{1}{|\mathbb{K}|^t} \cdot \frac{1}{(|\mathbb{K}|)_p} \cdot \frac{1}{|\mathbb{K}|}.
\end{aligned}$$

Conditioned on $\neg\mathsf{bad}_2$ in the good transcript, the definition of $\mathcal{L}$ implies that

$$\Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim \mathcal{L} \,\middle|\, \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}} \end{array}\right] = 1.$$

Then, $\neg\mathsf{bad}_1$ ensures that, for each $\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$, the permutation entries $\pi(\sigma(\mathsf{K}(\beta) - w_j))$ and $\pi^{-1}(-\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j))$ are not determined by other pairs in $\mathcal{Q}_{\mathcal{O}} \times [1, h-1]$. For those pairs in $(\mathcal{Q}_{\mathcal{O}} \times [1, h-1]) \setminus \mathcal{L}$, these two entries are even not determined by all queries in $\mathcal{Q}_\pi$ due to $\neg\mathsf{bad}_2$ and the definition of $\mathcal{L}$. By the randomness of π, we have

$$\Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim (\mathcal{Q}_{\mathcal{O}} \times [1, h-1]) \setminus \mathcal{L} \,\middle|\, \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_{\mathcal{O}} \wedge (\pi, \Delta) \sim \mathcal{L} \end{array}\right] = \frac{1}{(|\mathbb{K}| - p)_{t(h-1)-|\mathcal{L}|}}.$$

Using these results, we have

$$\begin{aligned}
\Pr_{\mathsf{real}}[\mathcal{Q}] &= \frac{1}{(|\mathbb{K}| - p)_{t(h-1)-|\mathcal{L}|} \cdot (|\mathbb{K}|)_p \cdot |\mathbb{K}|^{t+1}} \\
&= \frac{1}{(|\mathbb{K}|)_{p+t(h-1)-|\mathcal{L}|} \cdot |\mathbb{K}|^{t+1}}.
\end{aligned} \tag{7}$$

In the ideal world, it holds that

$$\begin{aligned}
\Pr_{\mathsf{ideal}}[\mathcal{Q}] &= \Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi \,\middle|\, \boldsymbol{c}(h) \sim \mathcal{Q}_{\mathcal{O}} \wedge \Delta\right] \cdot \Pr_{\mathsf{ideal}}[\boldsymbol{c}(h) \sim \mathcal{Q}_{\mathcal{O}} \wedge \Delta] \\
&= \Pr_{\mathsf{ideal}}\left[\begin{array}{c} \pi \sim \mathcal{Q}_\pi(\mathcal{L}) \\ \wedge\, \pi \sim \mathcal{Q}_\pi \setminus \mathcal{Q}_\pi(\mathcal{L}) \end{array} \,\middle|\, \boldsymbol{c}(h) \sim \mathcal{Q}_{\mathcal{O}} \wedge \Delta\right] \\
&\quad \cdot \Pr_{\mathsf{ideal}}[\boldsymbol{c}(h) \sim \mathcal{Q}_{\mathcal{O}}] \cdot \Pr_{\mathsf{ideal}}[\Delta] \\
&= \Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi \setminus \mathcal{Q}_\pi(\mathcal{L}) \,\middle|\, \begin{array}{c} \boldsymbol{c}(h) \sim \mathcal{Q}_{\mathcal{O}} \wedge \Delta \\ \wedge\, \pi \sim \mathcal{Q}_\pi(\mathcal{L}) \end{array}\right] \\
&\quad \cdot \Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi(\mathcal{L}) \,\middle|\, \boldsymbol{c}(h) \sim \mathcal{Q}_{\mathcal{O}} \wedge \Delta\right] \cdot \frac{1}{|\mathbb{K}|^{th} \cdot |\mathbb{K}|}.
\end{aligned}$$

By the programmed π and π^{-1} conditioned on $\neg\mathsf{bad}_2$, the defined $\mathcal{Q}_\pi(\mathcal{L})$ implies that

$$\Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi(\mathcal{L}) \,\middle|\, \boldsymbol{c}(h) \sim \mathcal{Q}_{\mathcal{O}} \wedge \Delta\right] = 1,$$

$$\Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi \setminus \mathcal{Q}_\pi(\mathcal{L}) \,\middle|\, \begin{array}{c} \boldsymbol{c}(h) \sim \mathcal{Q}_{\mathcal{O}} \wedge \Delta \\ \wedge\, \pi \sim \mathcal{Q}_\pi(\mathcal{L}) \end{array}\right] \leq \frac{1}{(|\mathbb{K}| - |\mathcal{Q}_\pi(\mathcal{L})|)_{p-|\mathcal{Q}_\pi(\mathcal{L})|}} = \frac{1}{(|\mathbb{K}| - |\mathcal{L}|)_{p-|\mathcal{L}|}}.$$

Using $|\mathcal{L}| \leq t(h-1)$, the above results, and (7), we have

$$\begin{aligned}\Pr_{\mathsf{ideal}}[\mathcal{Q}] &\leq \frac{1}{(|\mathbb{K}|-|\mathcal{L}|)_{p-|\mathcal{L}|}} \cdot \frac{1}{|\mathbb{K}|^{th} \cdot |\mathbb{K}|} \\ &\leq \frac{1}{(|\mathbb{K}|-t(h-1))_{p-|\mathcal{L}|}} \cdot \frac{1}{(|\mathbb{K}|)_{t(h-1)} \cdot |\mathbb{K}|^{t+1}} \\ &= \frac{1}{(|\mathbb{K}|)_{p+t(h-1)-|\mathcal{L}|} \cdot |\mathbb{K}|^{t+1}} = \Pr_{\mathsf{real}}[\mathcal{Q}]\,. \end{aligned} \tag{8}$$

From (6) and (8), we can have $\epsilon_1 = \mathsf{negl}(\lambda)$ and $\epsilon_2 = 0$. By the H-coefficient technique, $\mathcal{Z}$ can only distinguish the two worlds with a negligible advantage $\epsilon_1 + \epsilon_2$ if it is only given the transcript $\mathcal{Q} = (\Delta, \mathcal{Q}_\pi, \mathcal{Q}_\mathcal{O})$. This is exactly the advantage that $\mathcal{Z}$ can distinguish the two worlds since the output $\mathsf{K}(x)$'s in all **Extend** executions (the remaining part of the joint distribution) must be consistent with $\mathcal{Q}$ (or rather, its defined Δ and the receiver's cGGM tree) in the real world (due to the correctness of $\Pi_{\mathsf{sPCG-sVOLE}}$) and the ideal world. Therefore, this lemma holds. □

3 Proof of Theorem 2

Theorem 2. *In the $\mathcal{F}_{\mathsf{sVOLE}}$-hybrid model and under the $\mathsf{EA\text{-}LPN}(t, n, N, \ell, \mathbb{F})$ assumption, the protocol $\Pi_{\mathsf{sPCG\text{-}sVOLE}}$ (Fig. 5, but removing the consistency check) UC-realizes the functionality $\mathcal{F}_{\mathsf{sPCG\text{-}sVOLE}}$ (Fig. 4) against any semi-honest adversary, if hash function $\mathsf{H}(x) = \pi(\sigma(x)) + \sigma(x)$, where $\pi : \mathbb{K} \to \mathbb{K}$ is a (programmable) random permutation and $\sigma : \mathbb{K} \to \mathbb{K}$ is an efficient linear orthomorphism.*

Proof. **Corrupted sender.** Without the consistency check, the view of the corrupted sender contains only $\{\boldsymbol{d}_i\}_i$ and the outputs of $\mathcal{F}_{\mathsf{sVOLE}}$. These messages can be trivially simulated. Moreover, the output distribution of the sender can be consistent with the honest receiver due to the correctness of $\Pi_{\mathsf{sPCG\text{-}sVOLE}}$. Thus, the semi-honest security against the sender holds. Note that in this case, the sender has no chance (i.e., the consistency check) to learn the (on-average) one-bit leakage of the noise vector $\boldsymbol{e}$ so that the assumption $\mathsf{EA\text{-}LPN}(t, n, N, \ell, \mathcal{R})$ suffices.

Corrupted receiver. For the corrupted receiver, its view contains $\{\boldsymbol{c}_i\}_i$ and the outputs of $\mathcal{F}_{\mathsf{sVOLE}}$ (and the random permutation for H calls in cGGM). The simulation in this case is exactly that proved in the semi-honest case of [GYW+23, Theorem 7, full version], implying that the semi-honest security against the receiver. The simulator can be seen as a simplified case of the malicious one in the proof of Lemma 3 by removing the consistency check part and the emulation of random oracle RO. Its resulting proof and probability analysis are similar. The detailed proof is as follows.

When the receiver is corrupted, the simulator $\mathcal{S}$ in the ideal world emulates the random permutation π and its inverse π^{-1} throughout the lifespan of the distinguishing game. During the distinguishing game in either world, let $\mathcal{Q}_\pi$ denote an *ordered* list of the adversary's query-answer pairs of the random permutation or its inverse (note that $(x, y) \in \mathcal{Q}_\pi$ means that the adversary learns $\pi(x) = y$, regardless of whether it queried $\pi(x)$ or $\pi^{-1}(y)$), and $\mathcal{Q}_\mathcal{O}$ denote an *ordered* list of the adversary's query-answer pairs of the "oracle" emulated by the non-corrupted machine(s) (i.e., the honest P_0 plus the subroutine $\mathcal{F}_{\mathsf{sVOLE}}$ in the real world, or the simulator $\mathcal{S}$ in the ideal world). One can think that, for each $i \in [t]$ execution in the **Initialize** phase, $\mathcal{Q}_\mathcal{O}$ records the query $(\beta_i, \mathsf{M}(\beta_i), \boldsymbol{r}_i, \mathsf{M}(\boldsymbol{r}_i))$ (i.e., the effective sVOLE output committed by the corrupted P_1, see below) with its answer $\boldsymbol{c}_i$ (i.e., the transcript sent by the honest P_0). $\mathcal{S}$ has access to $\mathcal{Q}_\pi$ and $\mathcal{Q}_\mathcal{O}$ in the ideal world.

$\mathcal{S}$ sends (init) to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ upon receiving the first (init) from $\mathcal{A}$ to $\mathcal{F}_{\mathsf{sVOLE}}$, and works as follows for each $i \in [t]$ parallel execution in the **Initialize** phase:

1. Upon receiving $(\mathsf{extend}, h+1)$ from $\mathcal{A}$ to $\mathcal{F}_{\mathsf{sVOLE}}$, $\mathcal{S}$ waits for $\mathcal{A}$ to choose its sVOLE output $(\boldsymbol{s}_i, \mathsf{M}(\boldsymbol{s}_i))$.
2. $\mathcal{S}$ extracts $(\beta_i, \mathsf{M}(\beta_i))$, receives $\boldsymbol{d}_i$ from $\mathcal{A}$, and
    (a) If $\beta_i = 0$, picks a random $\boldsymbol{r}_i$.

(b) If $\beta_i \neq 0$, extracts $\boldsymbol{r}_i := \beta_i^{-1} \cdot (\boldsymbol{d}_i - \boldsymbol{s}_i{}^{(1,h)})$.

3. $\mathcal{S}$ computes $\mathsf{M}(\boldsymbol{r}_i) := \boldsymbol{r}_i \cdot \mathsf{M}(\beta_i) - \mathsf{M}(\boldsymbol{s}_i{}^{(1,h)})$.

4. $\mathcal{S}$ sends random $\boldsymbol{c}_i$ to $\mathcal{A}$ and appends $((\beta_i, \mathsf{M}(\beta_i), \boldsymbol{r}_i, \mathsf{M}(\boldsymbol{r}_i)), \boldsymbol{c}_i)$ to $\mathcal{Q}_{\mathcal{O}}$.

In the **Extend** phase:

1. $\mathcal{S}$ uses the same randomness to sample $\boldsymbol{b}$ and computes $(x, \mathsf{M}(x))$ as an honest P_1 in the protocol.

2. $\mathcal{S}$ sends (extend) and $(x, \mathsf{M}(x))$ to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$.

Moreover, $\mathcal{S}$ invokes **Global-key queries** to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ in the following cases (where, in $\mathcal{Q}_{\mathcal{O}}$, we omit the subscript of the parallel executions for clarity):

- query_1: $\mathcal{Z}$ queries the random permutation π with x or its inverse π^{-1} with y. For each $((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}) \in \mathcal{Q}_{\mathcal{O}}$ where $\beta \neq 0$:

  1. $\mathcal{S}$ computes
$$\begin{aligned} \alpha = \alpha_0 \dots \alpha_{h-1} &:= \overline{\boldsymbol{r}}^{(0)} \dots \overline{\boldsymbol{r}}^{(h-1)}, \\ \{z_i\}_{i\in[h]} &:= \mathsf{cGGM.OffPath}(\alpha, \{\mathsf{sum}^{(i)}_{\overline{\alpha}_i}\}_{i\in[h]}), \\ \forall j \in [1, h-1] : w_j &:= \textstyle\sum_{i\in[j]} z_i, \end{aligned} \tag{9}$$
     where $\mathsf{cGGM.OffPath}$ is a macro such that, on input a path $\alpha \in \{0,1\}^h$ and h sums $\{\mathsf{sum}^{(i)}_{\overline{\alpha}_i}\}_{i\in[h]}$ used in $\mathsf{cGGM.PuncFullEval}$ to define an h-level correlated GGM tree except the h on-path nodes, it outputs the h off-path nodes for the path α. This macro ensures that, for $j \in [h]$,
$$\begin{aligned} z_j = \mathsf{sum}^{(j)}_{\overline{\alpha}_j} - \text{“other } \overline{\alpha}_j\text{-side nodes on the } j\text{-th level} \\ \text{defined by the off-path nodes } \{z_i\}_{i\in[j]}\text{”}. \end{aligned}$$

  2. For each $j \in [1, h-1]$, $\mathcal{S}$ extracts $\mathsf{K}(\beta)'_1 \in \mathbb{K}$ and sends $(\mathsf{guess}, \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta)'_1))$ to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ if $\mathcal{Z}$ queries π with x, or extracts $\mathsf{K}(\beta)'_2 \in \mathbb{K}$ and $(\mathsf{guess}, \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta)'_2))$ to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ if $\mathcal{Z}$ queries π^{-1} with y, where
$$\begin{aligned} \mathsf{K}(\beta)'_1 &:= \sigma^{-1}(x) + w_j, \\ \mathsf{K}(\beta)'_2 &:= \begin{cases} \sigma^{-1}(-y + z_j) + w_j & \text{, if } \overline{\alpha}_j = 0 \\ \sigma'^{-1}(-y - z_j) + w_j & \text{, if } \overline{\alpha}_j = 1 \end{cases} \end{aligned} \tag{10}$$
     Note that $\sigma'(x) := \sigma(x) - x$ is a permutation as $\sigma : \mathbb{K} \to \mathbb{K}$ is an orthomorphism, and its inverse σ'^{-1} should be well-defined.

- query_2: A new pair $((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c})$ where $\beta \neq 0$ is added to $\mathcal{Q}_{\mathcal{O}}$ during execution.
  Using this pair, $\mathcal{S}$ computes $\{z_i\}_{i\in[h]}$ and $\{w_j\}_{j\in[1,h-1]}$ as per (9). Then, for each $(x, y) \in \mathcal{Q}_\pi$ and each $j \in [h]$, $\mathcal{S}$ extracts both $\mathsf{K}(\beta)'_1$ and $\mathsf{K}(\beta)'_2$ as above, and sends two global-key guesses $(\mathsf{guess}, \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta)'_1))$ and $(\mathsf{guess}, \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta)'_2))$ to $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$.

In either case, if $\mathcal{S}$ receives $(\mathsf{success})$ from $\mathcal{F}_{\mathsf{sPCG-sVOLE}}$ for some guess Δ, $\mathcal{S}$ will program the random permutation π and its inverse π^{-1} such that, for the offset $\mathsf{K}(\beta) := \mathsf{M}(\beta) - \beta \cdot \Delta$, the $\{z_i\}_{i\in[h]}$ and $\{w_j\}_{j\in[1,h-1]}$ computed from each pair in $\mathcal{Q}_{\mathcal{O}}$ up to this time, and each $j \in [1, h-1]$,
$$\pi(\sigma(\mathsf{K}(\beta) - w_j)) = -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j), \tag{11}$$
which must hold in the real world due to the construction of H and the definition of z_j and w_j. That is, the programming artificially ensures the real-world consistency among the off-path nodes on the cGGM tree in the ideal world.

We use H-coefficient technique to prove that this ideal world is statistically indistinguishable from the real one. Without loss of generality, we consider a deterministic environment $\mathcal{Z}$ and the transcript $\mathcal{Q}$, which is a part of the joint distribution of the view of $\mathcal{A}$ and the output of the honest P_0 (note that the remaining part of this joint distribution is the $\mathsf{K}(x)$'s in all **Extend** executions, and its consistency with $\mathcal{Q}$ will be checked in the end).

A transcript $\mathcal{Q} = (\Delta, \mathcal{Q}_\pi, \mathcal{Q}_{\mathcal{O}})$ is **bad** if it falls into one of the following cases:

- bad_1. There exist two distinct transcript pairs

$$\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1],$$
$$\Big(((\beta', \mathsf{M}(\beta'), \boldsymbol{r}', \mathsf{M}(\boldsymbol{r}')), \boldsymbol{c}'), j'\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$$

such that, for $\mathsf{K}(\beta) := \mathsf{M}(\beta) - \beta \cdot \Delta$ and $\mathsf{K}(\beta') := \mathsf{M}(\beta') - \beta' \cdot \Delta$, it holds that

$$\begin{aligned} &\Big(\mathsf{M}(\beta) - \beta \cdot \Delta - w_j = \mathsf{M}(\beta') - \beta' \cdot \Delta - w'_{j'}\Big) \\ \vee &\left(\begin{aligned} &-\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j) \\ &\quad = -\sigma(\mathsf{K}(\beta') - w'_{j'}) + (-1)^{\overline{\alpha}'_{j'}} \cdot (-\overline{\alpha}'_{j'} \cdot (\mathsf{K}(\beta') - w'_{j'}) + z'_{j'}) \end{aligned} \right), \end{aligned}$$

where the values are defined as per (9). This bad case captures the collision between the queries to the random permutation and its inverse.

- bad_2. There exists a transcript pair

$$\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$$

such that, for $\mathsf{K}(\beta) := \mathsf{M}(\beta) - \beta \cdot \Delta$, it holds that

$$\begin{aligned} &\left(\begin{aligned} &(\sigma(\mathsf{K}(\beta) - w_j), \dots) \in \mathcal{Q}_\pi \\ \vee\ &(\dots, -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)) \in \mathcal{Q}_\pi \end{aligned} \right) \\ \wedge &\Big(\pi(\sigma(\mathsf{K}(\beta) - w_j)) \neq -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j) \Big), \end{aligned}$$

where the values are defined as per (9). This bad case captures the inconsistency in the already defined entries of the random permutation and its inverse.

Consider bad_1 in the ideal world. For any two distinct pairs in $\mathcal{Q}_{\mathcal{O}} \times [1, h-1]$,

$$\begin{aligned} &\Pr_{\mathsf{ideal}} \left[\begin{aligned} &\Big(\mathsf{M}(\beta) - \beta \cdot \Delta - w_j = \mathsf{M}(\beta') - \beta' \cdot \Delta - w'_{j'}\Big) \\ \vee &\left(\begin{aligned} &-\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j) \\ &\quad = -\sigma(\mathsf{K}(\beta') - w'_{j'}) + (-1)^{\overline{\alpha}'_{j'}} \cdot (-\overline{\alpha}'_{j'} \cdot (\mathsf{K}(\beta') - w'_{j'}) + z'_{j'}) \end{aligned} \right) \end{aligned} \right] \\ &\le \Pr_{\mathsf{ideal}} \left[\mathsf{M}(\beta) - \beta \cdot \Delta - w_j = \mathsf{M}(\beta') - \beta' \cdot \Delta - w'_{j'} \right] \\ &\quad + \Pr_{\mathsf{ideal}} \left[\begin{aligned} &-\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j) \\ &\quad = -\sigma(\mathsf{K}(\beta') - w'_{j'}) + (-1)^{\overline{\alpha}'_{j'}} \cdot (-\overline{\alpha}'_{j'} \cdot (\mathsf{K}(\beta') - w'_{j'}) + z'_{j'}) \end{aligned} \right] \\ &= \frac{1}{|\mathbb{K}|} + \frac{1}{|\mathbb{K}|} = \frac{2}{|\mathbb{K}|}, \end{aligned}$$

where the probabilities come from that (i) the value of $\boldsymbol{c}$ in $\mathcal{Q}_{\mathcal{O}}$ is uniform and pairwise independent in the ideal world, and (ii) each z_j computed from $\boldsymbol{c}$ is also uniform due to (9). Taking a union bound over all distinct pairs over $\mathcal{Q}_{\mathcal{O}} \times [1, h-1]$, it holds that

$$\Pr_{\mathsf{ideal}} [\mathsf{bad}_1] \le \frac{t^2(h-1)^2}{|\mathbb{K}|}. \tag{12}$$

As for bad_2, it is sufficient to bound $\Pr_{\mathsf{ideal}} \left[\mathsf{bad}_2 \;\middle|\; \neg \mathsf{bad}_1 \right]$ using the following three sub-cases. In the *first* sub-case where

$$\begin{aligned} &\forall \Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1] : \\ &\quad \left(\begin{aligned} &(\sigma(\mathsf{K}(\beta) - w_j), \dots) \notin \mathcal{Q}_\pi \\ \wedge\ &(\dots, -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)) \notin \mathcal{Q}_\pi \end{aligned} \right), \end{aligned}$$

it is clear that bad_2 will not happen.

Otherwise, in the following two complement sub-cases, $\mathcal{S}$ can extract the global key Δ (which is a part of the transcript $\mathcal{Q}$) by solving $\mathsf{K}(\beta)$ from either $x = \sigma(\mathsf{K}(\beta) - w_j)$ or $y = -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)$ according to (10) and computing $\Delta = \beta^{-1} \cdot (\mathsf{M}(\beta) - \mathsf{K}(\beta))$ for each pair in $\mathcal{Q}_{\mathcal{O}}$ with $\beta \neq 0$. Then, $\mathcal{S}$ will use Δ to program π and π^{-1} to keep (11). If the programming succeeds, bad_2 will not happen since (11) holds for each pair in $\mathcal{Q}_{\mathcal{O}}$ and each $j \in [1, h-1]$.

In the *second* sub-case where a new query to π or π^{-1} triggers the programming (i.e., query_1 case), the programming must succeed conditioned on $\neg\mathsf{bad}_1$. The condition $\neg\mathsf{bad}_1$ ensures that, for every $\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$, the pre-image $x := \sigma(\mathsf{K}(\beta) - w_j)$ and the image $y := -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)$ to be programmed do not incur collision. More importantly, these two entries are not determined by the query/answer pairs in $\mathcal{Q}_\pi$ up to this time. Otherwise, $\exists (x, \dots) \in \mathcal{Q}_\pi$ or $\exists (\dots, y) \in \mathcal{Q}_\pi$ ahead of the current query such that the programming must have happened for the same $\mathcal{Q}_{\mathcal{O}}$ (i.e., contradicting that the programming is triggered by the current query). Hence, all pairs in $\mathcal{Q}_{\mathcal{O}}$ can be programmed as per (11). The answer to the current query is determined by the programmed result. The surely successful programming makes bad_2 impossible.

Consider the *third* sub-case where the programming arises from a new pair added to $\mathcal{Q}_{\mathcal{O}}$ (i.e., query_2 case). Assume that the global key Δ (and its associated $\mathsf{K}(\beta) = \mathsf{M}(\beta) - \beta \cdot \Delta$) to be used in the programming is extracted from this pair, an existing $(x^*, \dots) \in \mathcal{Q}_\pi$ or $(\dots, y^*) \in \mathcal{Q}_\pi$, and $j^* \in [1, h-1]$. Note that either $(x^*, \dots) \in \mathcal{Q}_\pi$ or $(\dots, y^*) \in \mathcal{Q}_\pi$ has been given to $\mathcal{Z}$ and *cannot* be programmed. We bound the probability of this *undesired* sub-case. In the ideal world,

$$
\begin{aligned}
&\Pr_{\mathsf{ideal}}\left[\begin{array}{r} (x^* = \sigma(\mathsf{K}(\beta) - w_{j^*})) \\ \vee\, (y^* = -\sigma(\mathsf{K}(\beta) - w_{j^*}) + (-1)^{\overline{\alpha}_{j^*}} \cdot (-\overline{\alpha}_{j^*} \cdot (\mathsf{K}(\beta) - w_{j^*}) + z_{j^*})) \end{array}\right] \\
&\leq \Pr_{\mathsf{ideal}}\left[x^* = \sigma(\mathsf{K}(\beta) - w_{j^*})\right] \\
&\qquad + \Pr_{\mathsf{ideal}}\left[y^* = -\sigma(\mathsf{K}(\beta) - w_{j^*}) + (-1)^{\overline{\alpha}_{j^*}} \cdot (-\overline{\alpha}_{j^*} \cdot (\mathsf{K}(\beta) - w_{j^*}) + z_{j^*})\right] \\
&= \Pr_{\mathsf{ideal}}\left[w_{j^*} = \mathsf{K}(\beta) - \sigma^{-1}(x^*)\right] \\
&\qquad + \Pr_{\mathsf{ideal}}\left[z_{j^*} = (-1)^{\overline{\alpha}_{j^*}} \cdot \Big(y^* + \sigma(\mathsf{K}(\beta) - w_{j^*})\Big) + \overline{\alpha}_{j^*} \cdot (\mathsf{K}(\beta) - w_{j^*})\right] \\
&= \frac{1}{|\mathbb{K}|} + \frac{1}{|\mathbb{K}|} = \frac{2}{|\mathbb{K}|},
\end{aligned}
$$

where the values are computed from the newly added pair as per (9), and the probabilities are taken over the uniform $\boldsymbol{c}$ in $\mathcal{Q}_{\mathcal{O}}$. Taking a union bound over $\mathcal{Q}_\pi \times \mathcal{Q}_{\mathcal{O}} \times [1, h-1]$ (where we assume $|\mathcal{Q}_\pi| = p = p(\lambda)$ is polynomially bounded), we can see that this sub-case happens with probability at most $2pt(h-1)/\left|\mathbb{K}\right|$.

Combining the above three sub-cases, we have that

$$
\Pr_{\mathsf{ideal}}\left[\mathsf{bad}_2 \;\middle|\; \neg\,\mathsf{bad}_1\right] \leq \frac{2pt(h-1)}{|\mathbb{K}|}. \tag{13}
$$

Using (12) and (13), it holds that

$$
\begin{aligned}
\Pr_{\mathsf{ideal}}\left[\mathcal{Q} \in \mathcal{T}_{\mathsf{bad}}\right] &= \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1 \vee \mathsf{bad}_2\right] \\
&= \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1\right] + \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_2\right] - \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1 \wedge \mathsf{bad}_2\right] \\
&= \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1\right] + \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_2 \;\middle|\; \neg\,\mathsf{bad}_1\right] \cdot \Pr_{\mathsf{ideal}}\left[\neg\mathsf{bad}_1\right] \\
&\leq \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_1\right] + \Pr_{\mathsf{ideal}}\left[\mathsf{bad}_2 \;\middle|\; \neg\,\mathsf{bad}_1\right] \\
&\leq \frac{t^2(h-1)^2 + 2pt(h-1)}{|\mathbb{K}|} = \mathsf{negl}(\lambda).
\end{aligned} \tag{14}
$$

We proceed to bound the ratio of $\Pr_{\mathsf{real}}\left[\mathcal{Q}\right]$ to $\Pr_{\mathsf{ideal}}\left[\mathcal{Q}\right]$ for some fixed **good** transcript $\mathcal{Q} =$

$(\Delta, \mathcal{Q}_\pi, \mathcal{Q}_\mathcal{O})$. Let $\mathcal{L} \subseteq \mathcal{Q}_\mathcal{O} \times [1, h-1]$ such that

$$\forall\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{L} :$$
$$\begin{pmatrix} & (\sigma(\mathsf{K}(\beta) - w_j), \dots) \in \mathcal{Q}_\pi \\ \vee\, (\dots, -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)) \in \mathcal{Q}_\pi \end{pmatrix}.$$

Conditioned on $\neg(\mathsf{bad}_1 \vee \mathsf{bad}_2)$ implicit in good transcripts, $\mathcal{L}$ induces a subset $\mathcal{Q}_\pi(\mathcal{L}) \subseteq \mathcal{Q}_\pi$. There exists a *bijective* correspondence between a $(x, y) \in \mathcal{Q}_\pi(\mathcal{L})$ and the (z_j, w_j) computed as per (9) from a pair in $\mathcal{L}$ such that

$$x = \sigma(\mathsf{K}(\beta) - w_j),$$
$$y = -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j).$$

It is clear that $|\mathcal{L}| = |\mathcal{Q}_\pi(\mathcal{L})| \le \min(|\mathcal{Q}_\pi|, |\mathcal{Q}_\mathcal{O}| \cdot (h-1)) = \min(p, t(h-1))$.

For $\mathcal{X} \subseteq \mathcal{Q}_\pi$, let $\pi \sim \mathcal{X}$ be the event that the permutation π is consistent with the queries and answers in $\mathcal{X}$, i.e., $\forall (x, y) \in \mathcal{X} : \pi(x) = y$. Let $\boldsymbol{c}(j) \sim \mathcal{Q}_\mathcal{O}$ be the event that the prefix $(\boldsymbol{c}_i{}^{(0)}, \dots, \boldsymbol{c}_i{}^{(j-1)})$ in each $i \in [t]$ parallel execution in the **Initialize** phase take the literal values in $\mathcal{Q}_\mathcal{O}$. For $\mathcal{Y} \subseteq \mathcal{Q}_\mathcal{O} \times [1, h-1]$, let $(\pi, \Delta) \sim \mathcal{Y}$ be the *real-world* event that π and the global key Δ are consistent with the pairs in $\mathcal{Y}$, i.e.,

$$\forall\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Y} :$$
$$\pi(\sigma(\mathsf{K}(\beta) - w_j)) = -\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j)$$

for z_j and w_j computed from (9). Let $(a)_b := a \cdots (a - b + 1)$ be falling factorial.

In the real world, $\pi : \mathbb{K} \to \mathbb{K}$ is a random permutation without programming, and

$$\begin{aligned}
\Pr_{\mathsf{real}}[\mathcal{Q}] &= \Pr_{\mathsf{real}}\big[\boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O} \wedge (\pi, \Delta) \sim \mathcal{Q}_\mathcal{O} \times [1, h-1] \;\big|\; \pi \sim \mathcal{Q}_\pi \wedge \Delta\big] \\
&\quad \cdot \Pr_{\mathsf{real}}[\pi \sim \mathcal{Q}_\pi \wedge \Delta] \\
&= \Pr_{\mathsf{real}}\big[\boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O} \wedge (\pi, \Delta) \sim \mathcal{Q}_\mathcal{O} \times [1, h-1] \;\big|\; \pi \sim \mathcal{Q}_\pi \wedge \Delta\big] \\
&\quad \cdot \Pr_{\mathsf{real}}[\pi \sim \mathcal{Q}_\pi] \cdot \Pr_{\mathsf{real}}[\Delta] \\
&= \Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim \mathcal{Q}_\mathcal{O} \times [1, h-1] \;\middle|\; \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O} \end{array}\right] \\
&\quad \cdot \Pr_{\mathsf{real}}\big[\boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O} \;\big|\; \pi \sim \mathcal{Q}_\pi \wedge \Delta\big] \cdot \Pr_{\mathsf{real}}[\pi \sim \mathcal{Q}_\pi] \cdot \Pr_{\mathsf{real}}[\Delta] \\
&= \Pr_{\mathsf{real}}\left[\begin{array}{l} (\pi, \Delta) \sim \mathcal{L} \\ \quad \wedge\, (\pi, \Delta) \sim (\mathcal{Q}_\mathcal{O} \times [1, h-1]) \setminus \mathcal{L} \end{array} \;\middle|\; \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O} \end{array}\right] \\
&\quad \cdot \Pr_{\mathsf{real}}[\boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O}] \cdot \Pr_{\mathsf{real}}[\pi \sim \mathcal{Q}_\pi] \cdot \Pr_{\mathsf{real}}[\Delta] \\
&= \Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim (\mathcal{Q}_\mathcal{O} \times [1, h-1]) \setminus \mathcal{L} \;\middle|\; \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O} \wedge (\pi, \Delta) \sim \mathcal{L} \end{array}\right] \\
&\quad \cdot \Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim \mathcal{L} \;\middle|\; \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O} \end{array}\right] \cdot \frac{1}{|\mathbb{K}|^t} \cdot \frac{1}{(|\mathbb{K}|)_p} \cdot \frac{1}{|\mathbb{K}|}.
\end{aligned}$$

Conditioned on $\neg\mathsf{bad}_2$ in the good transcript, the definition of $\mathcal{L}$ implies that

$$\Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim \mathcal{L} \;\middle|\; \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O} \end{array}\right] = 1.$$

Then, $\neg\mathsf{bad}_1$ ensures that, for each $\Big(((\beta, \mathsf{M}(\beta), \boldsymbol{r}, \mathsf{M}(\boldsymbol{r})), \boldsymbol{c}), j\Big) \in \mathcal{Q}_\mathcal{O} \times [1, h-1]$, the permutation entries $\pi(\sigma(\mathsf{K}(\beta) - w_j))$ and $\pi^{-1}(-\sigma(\mathsf{K}(\beta) - w_j) + (-1)^{\overline{\alpha}_j} \cdot (-\overline{\alpha}_j \cdot (\mathsf{K}(\beta) - w_j) + z_j))$ are not determined by other pairs in $\mathcal{Q}_\mathcal{O} \times [1, h-1]$. For those pairs in $(\mathcal{Q}_\mathcal{O} \times [1, h-1]) \setminus \mathcal{L}$, these two entries are even not determined by all queries in $\mathcal{Q}_\pi$ due to $\neg\mathsf{bad}_2$ and the definition of $\mathcal{L}$. By the randomness of π, we have

$$\Pr_{\mathsf{real}}\left[(\pi, \Delta) \sim (\mathcal{Q}_\mathcal{O} \times [1, h-1]) \setminus \mathcal{L} \;\middle|\; \begin{array}{c} \pi \sim \mathcal{Q}_\pi \wedge \Delta \\ \wedge\, \boldsymbol{c}(1) \sim \mathcal{Q}_\mathcal{O} \wedge (\pi, \Delta) \sim \mathcal{L} \end{array}\right] = \frac{1}{(|\mathbb{K}| - p)_{t(h-1) - |\mathcal{L}|}}.$$

Using these results, we have

$$\Pr_{\mathsf{real}}[\mathcal{Q}] = \frac{1}{(|\mathbb{K}|-p)_{t(h-1)-|\mathcal{L}|}\cdot(|\mathbb{K}|)_p\cdot|\mathbb{K}|^{t+1}} = \frac{1}{(|\mathbb{K}|)_{p+t(h-1)-|\mathcal{L}|}\cdot|\mathbb{K}|^{t+1}}. \tag{15}$$

In the ideal world, it holds that

$$\begin{aligned}\Pr_{\mathsf{ideal}}[\mathcal{Q}] &= \Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi \,\middle|\, \boldsymbol{c}(h) \sim \mathcal{Q}_\mathcal{O} \wedge \Delta\right] \cdot \Pr_{\mathsf{ideal}}[\boldsymbol{c}(h) \sim \mathcal{Q}_\mathcal{O} \wedge \Delta] \\ &= \Pr_{\mathsf{ideal}}\left[\begin{array}{c}\pi \sim \mathcal{Q}_\pi(\mathcal{L}) \\ \wedge\ \pi \sim \mathcal{Q}_\pi \setminus \mathcal{Q}_\pi(\mathcal{L})\end{array} \,\middle|\, \boldsymbol{c}(h) \sim \mathcal{Q}_\mathcal{O} \wedge \Delta\right] \\ &\quad \cdot \Pr_{\mathsf{ideal}}[\boldsymbol{c}(h) \sim \mathcal{Q}_\mathcal{O}] \cdot \Pr_{\mathsf{ideal}}[\Delta] \\ &= \Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi \setminus \mathcal{Q}_\pi(\mathcal{L}) \,\middle|\, \begin{array}{c}\boldsymbol{c}(h) \sim \mathcal{Q}_\mathcal{O} \wedge \Delta \\ \wedge\ \pi \sim \mathcal{Q}_\pi(\mathcal{L})\end{array}\right] \\ &\quad \cdot \Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi(\mathcal{L}) \,\middle|\, \boldsymbol{c}(h) \sim \mathcal{Q}_\mathcal{O} \wedge \Delta\right] \cdot \frac{1}{|\mathbb{K}|^{th}\cdot|\mathbb{K}|}.\end{aligned}$$

By the programmed π and π^{-1} conditioned on $\neg\mathsf{bad}_2$, the defined $\mathcal{Q}_\pi(\mathcal{L})$ implies that

$$\Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi(\mathcal{L}) \,\middle|\, \boldsymbol{c}(h) \sim \mathcal{Q}_\mathcal{O} \wedge \Delta\right] = 1,$$

$$\Pr_{\mathsf{ideal}}\left[\pi \sim \mathcal{Q}_\pi \setminus \mathcal{Q}_\pi(\mathcal{L}) \,\middle|\, \begin{array}{c}\boldsymbol{c}(h) \sim \mathcal{Q}_\mathcal{O} \wedge \Delta \\ \wedge\ \pi \sim \mathcal{Q}_\pi(\mathcal{L})\end{array}\right] \leq \frac{1}{(|\mathbb{K}|-|\mathcal{Q}_\pi(\mathcal{L})|)_{p-|\mathcal{Q}_\pi(\mathcal{L})|}} = \frac{1}{(|\mathbb{K}|-|\mathcal{L}|)_{p-|\mathcal{L}|}}.$$

Using $|\mathcal{L}| \leq t(h-1)$, the above results, and (15), we have

$$\begin{aligned}\Pr_{\mathsf{ideal}}[\mathcal{Q}] &\leq \frac{1}{(|\mathbb{K}|-|\mathcal{L}|)_{p-|\mathcal{L}|}} \cdot \frac{1}{|\mathbb{K}|^{th}\cdot|\mathbb{K}|} \\ &\leq \frac{1}{(|\mathbb{K}|-t(h-1))_{p-|\mathcal{L}|}} \cdot \frac{1}{(|\mathbb{K}|)_{t(h-1)}\cdot|\mathbb{K}|^{t+1}} \\ &= \frac{1}{(|\mathbb{K}|)_{p+t(h-1)-|\mathcal{L}|}\cdot|\mathbb{K}|^{t+1}} = \Pr_{\mathsf{real}}[\mathcal{Q}].\end{aligned} \tag{16}$$

From (14) and (16), we can have $\epsilon_1 = \mathsf{negl}(\lambda)$ and $\epsilon_2 = 0$. By the H-coefficient technique, $\mathcal{Z}$ can only distinguish the two worlds with a negligible advantage $\epsilon_1 + \epsilon_2$ if it is only given the transcript $\mathcal{Q} = (\Delta, \mathcal{Q}_\pi, \mathcal{Q}_\mathcal{O})$. This is exactly the advantage that $\mathcal{Z}$ can distinguish the two worlds since the output $\mathsf{K}(x)$'s in all **Extend** executions (the remaining part of the joint distribution) must be consistent with $\mathcal{Q}$ (or rather, its defined Δ and the receiver's cGGM tree) in the real world (due to the correctness of $\Pi_{\mathsf{sPCG-sVOLE}}$) and the ideal world. □

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